Questions

Q1.

Figure 17 shows a Geiger-Müller (GM) tube used for measuring radioactivity.

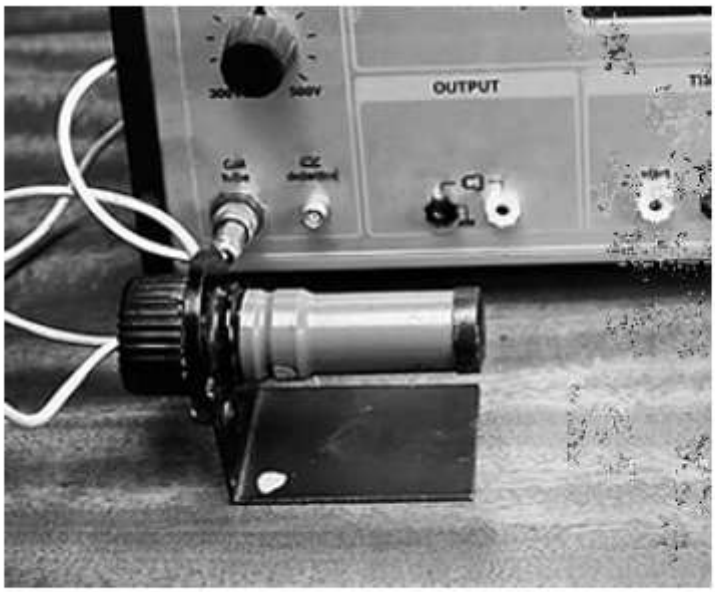

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Figure 17

A hospital uses a radioactive isotope with a half-life of 6 hours.

A technician measures a count rate of 80 counts per minute (cpm) from this isotope.

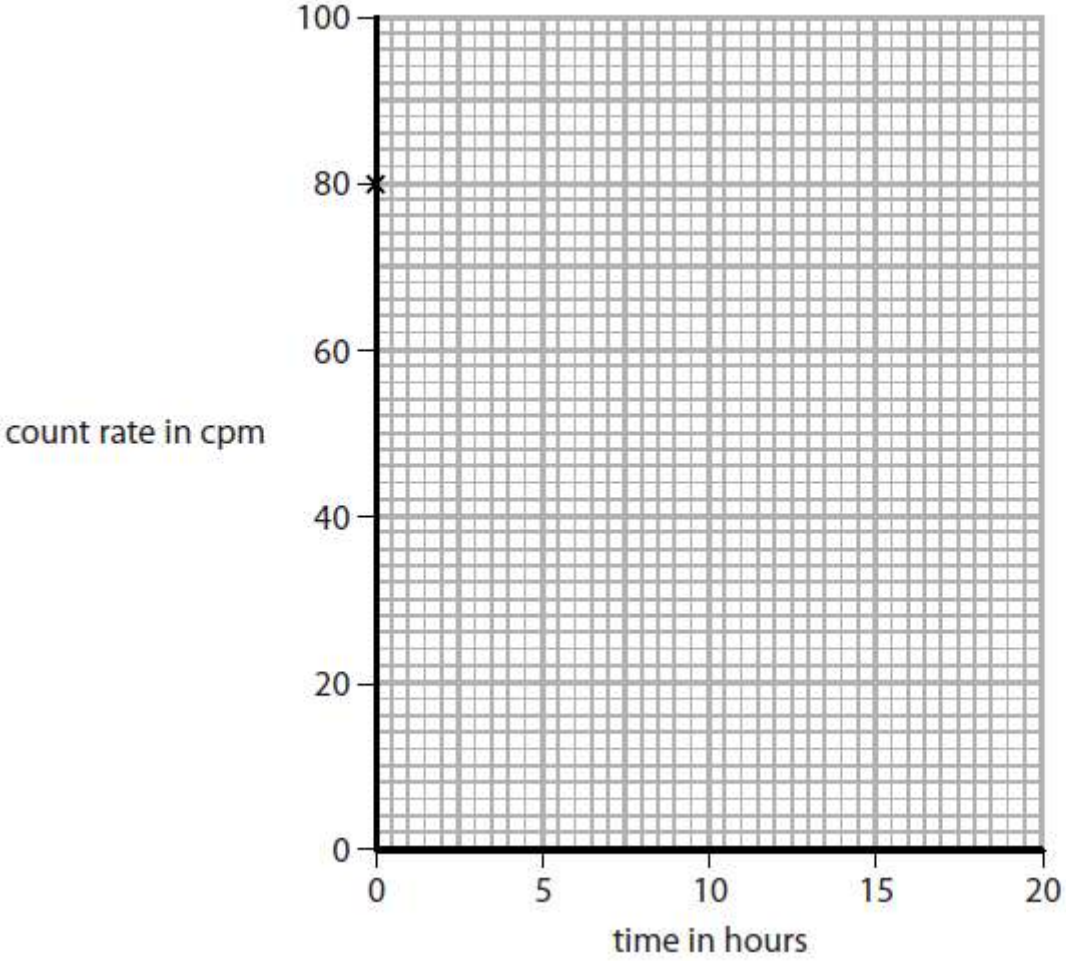


Figure 18

Complete the graph on Figure 18, as accurately as possible, to show how the count-rate from this isotope will change from the time of the first measurement.

The first point is already drawn in Figure 18.

(3)

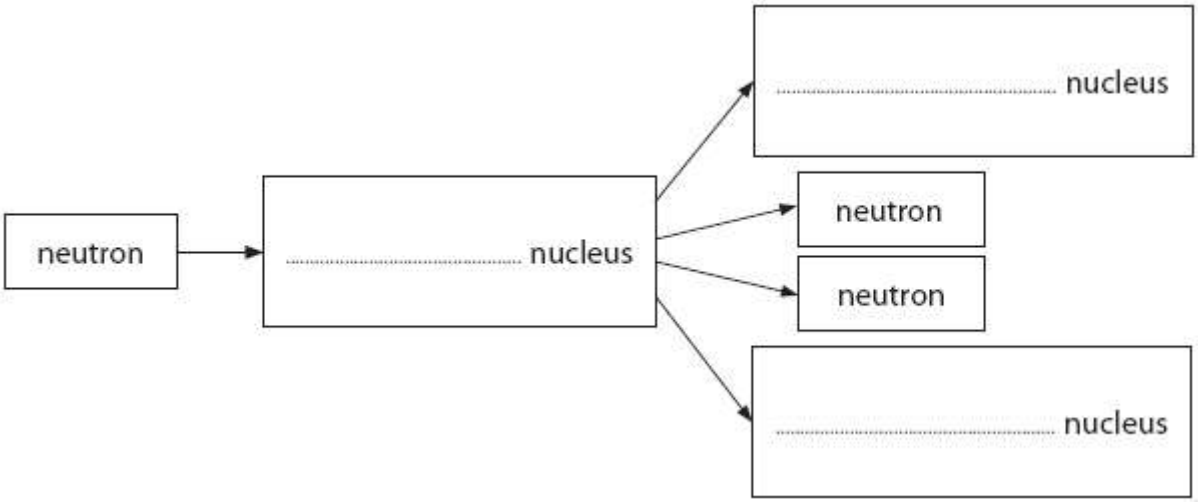
(Total for question = 3 marks)

Q2.

Beryllium-9 is a stable isotope of beryllium.
A beryllium-9 nucleus absorbs a neutron.
After a short time the new nucleus splits into two neutrons and two alpha particles.

(i) Complete the flow chart for this reaction.

(2)



(ii) Compare this nuclear reaction with the fission of a uranium nucleus.

(3)

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(iii) A fission reaction can be the start of a chain reaction.

Describe what needs to happen next to produce a chain reaction.

(2)

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Q3.

(a) An underground oil pipe starts to leak oil.
To find the leak, a technician adds a gamma source to the oil flowing in the pipe.

Describe how the technician can find the position of the leak.

(2)

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(b) Which of these is correct for half-life?

Put a cross (☒) in the box next to your answer.

(1)

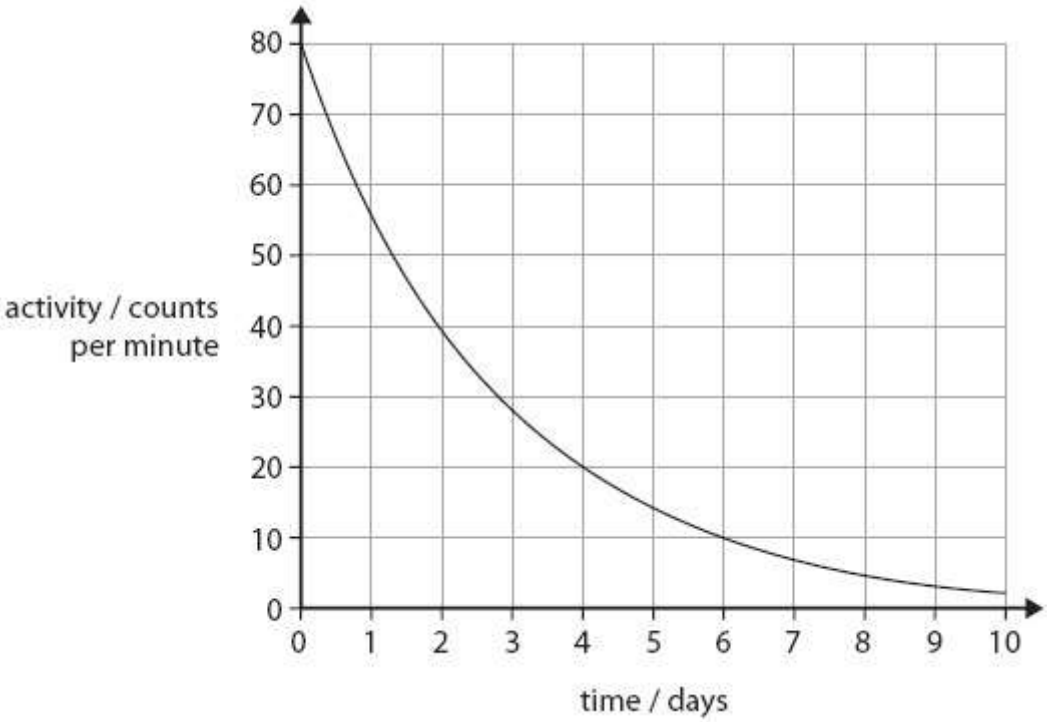
☒ **A** It is half the time for all the atoms to decay

☒ **B** It is the time it takes for an atom to half decay

☒ **C** It is the time it takes for half an atom to decay

☒ **D** It is the time it takes for half the atoms to decay

(c) The graph shows how the activity of a sample of a radioactive material changes with time.
The sample has an initial activity of 80 counts per minute.



(i) Use the graph to find the half-life of the material.

(1)

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(ii) Another sample of the material has an initial count rate of 40 counts per minute.

Sketch, on the same axes, the activity of this sample for the first 4 days.

(2)

*(d) Some scientists carry out an experiment to measure the radioactivity from a source to be used in a factory.
They measure the background radiation before and after their experiment.
They take the background count at the same place as they do their experiment.

Explain how this procedure helps to make sure that the results of the experiment are valid.

(6)

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(Total for Question = 12 marks)

Q4.

(a) Here are four uses of radioactivity.
Draw a line from each one of them to the type of radiation it uses.

Each type of radiation may be chosen once, more than once or not at all.

(4)

Use of radioactivity	Type of radiation it uses
sterilisation of medical equipment	alpha
household fire (smoke) alarm	beta
gauging thickness of cardboard	gamma
irradiating food	

(b) Which graph best shows how the activity of a radioactive isotope changes with time?

Put a cross ( ) in the box next to your answer.

(1)

(c) Complete the sentence by putting a cross ( ) in the box next to your answer.

The unit of activity of a radioactive isotope is the

(1)

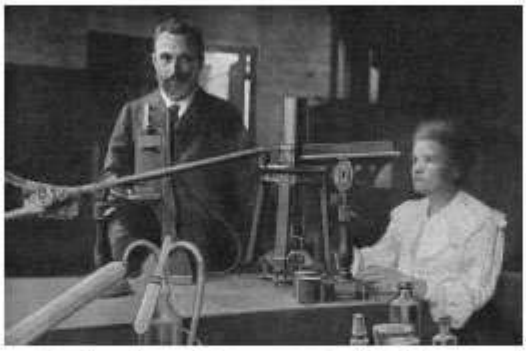
☒ **A** americium

☒ **B** becquerel

☒ **C** einstein

☒ **D** radium

Marie Curie investigated radioactivity over 100 years ago.



She often carried radioactive materials in her pocket.
She stored them in her desk drawer.
She liked the coloured light they gave off.
Marie probably died from exposure to their radiation.

(d)

Describe **two** precautions that scientists now take when they use radioactive materials.

(2)

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(Total for Question = 8 marks)

Q5.

(a) Cobalt-60 is a radioactive substance.
A nucleus of cobalt-60 contains 27 protons and 33 neutrons.

(i) Complete the sentence by putting a cross (☒) in the box next to your answer.
The number of electrons in a neutral atom of cobalt-60 is

(1)

☒ **A** 87
☒ **B** 60
☒ **C** 33
☒ **D** 27

(ii) Cobalt-60 decays by emitting gamma radiation.
Explain what happens to the mass of a cobalt-60 atom when a gamma ray is emitted.

(2)

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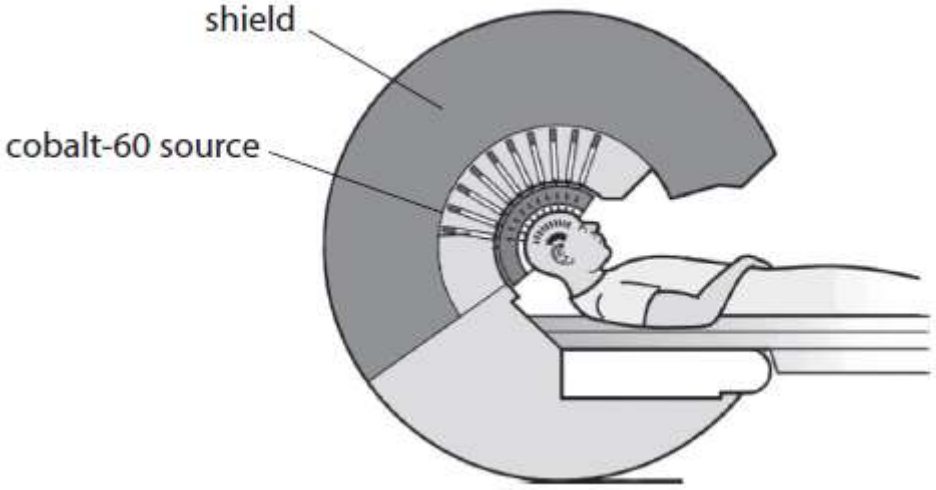
(b) Gamma radiation from cobalt-60 can be used to destroy tumours.
The diagram shows how gamma radiation is used to destroy a brain tumour.



(i) Complete the sentence by putting a cross (☒) in the box next to your answer.
Gamma radiation is used because

(1)

☐ **A** gamma can penetrate further than alpha or beta
☐ **B** gamma is more ionising than alpha or beta
☐ **C** gamma is always safer than alpha or beta
☐ **D** gamma has a shorter half-life than alpha or beta

(ii) Describe what the shield is used for.

(2)

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(iii) Suggest **two** advantages that this kind of treatment has over other forms of treatment for tumours.

(2)

1
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2
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(iv) Explain why several beams of gamma radiation are used instead of just one.

(2)

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(Total for Question = 10 marks)

Q6.

* Describe how radioactive materials can be used in the diagnosis and treatment of some illnesses.

(6)

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Q7.

Some radioactive isotopes emit positrons.

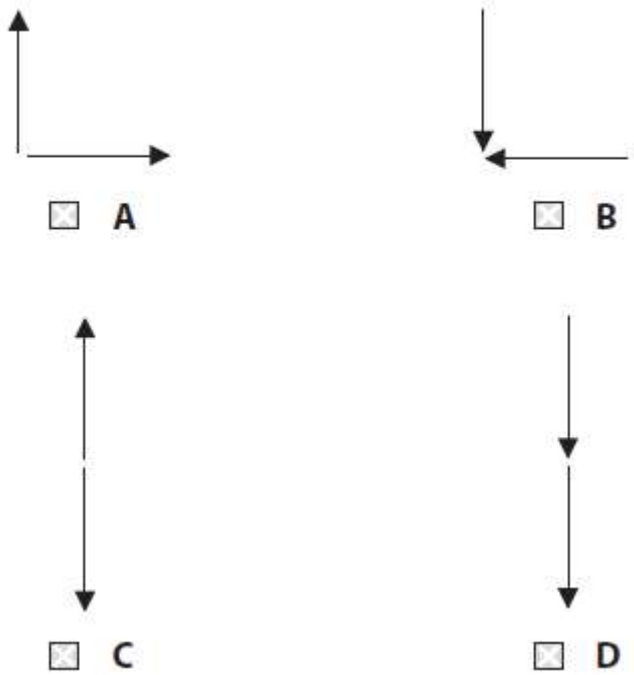
Positrons are used to make gamma rays.

When a positron annihilates an electron, two gamma rays are produced.

(i) Which diagram shows the directions of the two gamma rays produced?

Put a cross (☒) in the box next to your answer.

(1)



(ii) Explain how charge is conserved when an electron annihilates a positron.

(3)

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(iii) Explain how mass and energy are conserved when an electron annihilates a positron.

(2)

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Q8.

Hospitals use ionising radiation for many purposes.

(a) State **one** use of ionising radiation in a hospital.

(1)

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(b) An isotope of technicium, technicium-99, has a half-life of 6 hours.

A hospital has a sample which contains 40 mg of technicium-99.
Calculate how much technicium-99 will be in this sample after 12 hours.

(2)

amount remaining = ... mg

Q9.

Figure 8 shows a helium nucleus.

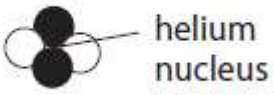

Figure 8

Two of the particles in the helium nucleus are neutrons.

State the name of the other two particles in the helium nucleus.

(1)

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(Total for question = 1 mark)

Q10.

There are many control rods in a nuclear reactor.

Explain how control rods are used to reduce the number of nuclear reactions in the reactor.

(2)

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Mark Scheme

Q1.

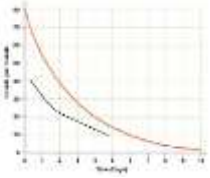
Question Number	Answer	Additional guidance	Mark
	• point after first half-life - 6, 40 (1) • point after second half-life - 12, 20 (1) • point after third half-life - 18, 10 (1)	within 1 small square by eye smooth curve starting at 80, with a decreasing gradient passing through one correct half-life point scores 2 marks smooth curve starting at 80, with a decreasing gradient passing through two correct half-life points scores 3 marks **if no other mark scored** smooth curve showing decreasing gradient but not going through any correct points scores 1 mark	**(3)** AO 3 1a

Q2.

	Answer	Acceptable answers	Mark
(i)	heli um beryllium (1) (1) he lium	daughter in right hand boxes daughter	**(2)**
(ii)	a comparison which describes any **three** of the following: similarities: • produce (more)	different elements / smaller nuclei for daughters do not accept split an atom neutron is	**(3)**

	neutrons (1) • produce 'daughter' (nuclei) (1) • release energy (1) • split a (bigger) nucleus (1) • (triggered by) a neutron coming in (1) • nucleus becomes unstable (before splitting) (1)differences: • uranium daughters are different from each other/ beryllium daughters are the same (1) • uranium daughters are heavier than beryllium daughters (1)	absorbed	
(iii)	a description including: neutron(s) (from first fission) (1) (go on to) cause another fission (1)	collide with another nucleus /atom	**(2)**

Q3.

	Answer	**Acceptable answers**	**Mark**
(a)	A description to include • name of detector / move detector over the ground (1) • where leak is, there will be an increased rate (1)	(move) until a leak/high reading is found	**(2)**
(b)	D It is the time it takes for half the atoms to decay		**(1)**
(c)i	1.9-2 (days)		**(1)**
(c)ii	 plotting (0,40), (2,20) and (4,10) **OR ANY** line which passes	Ignore any part of line after 4 days	**(2)**

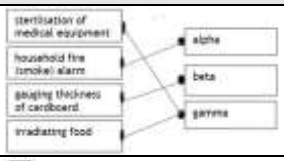
	through those coordinates (1) smooth curve through those points (1)		

		Indicative Content	Mark
QWC	***(d)**	An explanation including some of the following ideas **Need for measurement (N)**Background radiation • is always present/all around us • has (natural) source(s) exemplified by space, living things, rocks, food, nuclear/medical sources etc. • would give false reading in experiment **How and why to measure(H)** Background radiation measurement • is taken at site of experiment **because** it is different in different places • is taken with all apparatus **except** source in place • is taken before and after **because** it can change with time / they need an average • must be worked out for same time as (or longer than) experiment / rate found **so** analysis is simpler • It is taken several times/ averaged **because** it is random **Analysis (A)** Background radiation measurement • must be	**(6)**

		subtracted from measurements with source /main count rate	
Level	**0**	No rewardable content	
1	**1 - 2**	• A limited explanation mentioning any **two** from N or **one** from H or A e.g. Background comes from space and rocks.(N) It is there all the time. (N) OR Readings for background must be repeated because they are random. (H) OR Background must be taken away from all other readings (A) • the answer communicates ideas using simple language and uses limited scientific terminology • spelling, punctuation and grammar are used with limited accuracy	
2	**3 - 4**	• A simple explanation ***linking*** aspects of two ideas i.e. **N** + **H** OR **N** + **A** OR **H** + **A** e.g Take readings without source (H) and subtract them from the main readings with source present.(A) OR It should be taken several times because it is random (H)so that the average can be subtracted from the main readings (A) • the answer communicates ideas showing some evidence of clarity and organisation and uses scientific terminology appropriately • spelling, punctuation and grammar are used with some accuracy	
3	**5 - 6**	• A detailed explanation ***linking*** **A** with EITHER N + an idea from **H** OR two or more ideas from **H** e.g.Background radiation is there all the time. (N) You need to take readings at the place where you will do the experiment and with all the apparatus set up except the source because BR changes from place to place.(H) Then you should subtract background readings from the main experimental readings.(A) OR Take several readings of count rate for averaging since the effect is random (H) and make sure that they are taken in the same place.(H) Then subtract from readings in main experiment.(A) • the answer communicates ideas clearly and coherently uses a range of scientific terminology accurately	

		• spelling, punctuation and grammar are used with few errors

Q4.

	Answer	Acceptable answers	Mark
(a)		one mark for each correct line	**(4)**
(b)	☒ **D**		**(1)**
(c)	☒ **B** becquerel		**(1)**
(d)	A description including any **two** from: • secure storage (1): • avoid direct contact (1) • wear protective clothing (1) • minimise exposure (1) • shielding (1) • minimise dose (1) • monitor exposure (1) • protect other people (1)	either the purpose, such as to prevent radiation getting out or a description such as lead-lined box/locked away when not in use. do not touch / use tongs /wash after handling lead lined suits/aprons/masks/gloves ignore goggles long distance away / not pointing towards body/ keep sources shielded /stand behind shields short time wear film badge/use Geiger counter (to monitor radiation levels) warning signs / barriers / restricted areas /controlled areas	**(2)**

Q5.

Question Number	Answer	Acceptable answers	Mark
(a)(i)	**D** 27 (1)		**(1)**

Question Number	Answer	Acceptable answers	Mark
(a)(ii)	an explanation linking: • no change in mass (number) (1) • (because) gamma is a wave (electromagnetic) / has no mass (itself) (1) OR • mass decreases (1) • idea of mass – energy equivalence (1) (must be clearly stated)	gamma is only energy / not a particle nucleus de-excites / rearranged for one mark do not allow 'mass number decreases'	**(2)**

Question Number	Answer	Acceptable answers	Mark
(b) (i)	**A** gamma can penetrate further than alpha or beta (1)		**(1)**

Question Number	Answer	Acceptable answers	Mark
(b) (ii)	description to include: • protects / stops radiation escaping (1) • affecting operator/doctor/nurse (1)	absorbs (radiation) other people / others	**(2)**

Question Number	Answer	Acceptable answers	Mark
(b) (iii)	two from: • non invasive / no surgery required (1) • no radioactive substances left in the body (1) • no anaesthetic used • patient does not become radioactive (1) • outpatient procedure (1) • does not affect the whole body (1) • (accurate) targeting of tumour (1) • painless (at the time) for the patient • procedure (may be) quicker	no need to operate / cut open patient / reduces risk of infection no harmful side effects like chemotherapy ignore answers\that apply equally to other treatments e.g. 'kills cancer'	**(2)**

Question Number	Answer	Acceptable answers	Mark
(b) (iv)	explanation linking two from: • idea of targeting / beams concentrate / focus on tumour (1) • avoid damage to healthy cells / tissue (1) • (reaching / getting to) all parts of the tumour (1)	more rays hit tumour / beams overlap at tumour ignore '(more) beams penetrate more' / (more) accurate	**(2)**

(Total for Question = 10 marks)

Q6.

Question Number		Indicative Content	Mark
QWC		A description including some of the following points :- Diagnosis • radioactive tracers used in the body • to check systems e.g. skeleton / bone, blood flow, thyroid activity, kidney function. • attached to some compound which targets a particular area of the body • radioactive tracers are isotopes with short half-lives put into the body • may go into PET scans, since this involves beta+ emitters • gamma cameras, used to detect emissions from radioactive tracers Treatment • radiotherapy, use of gamma rays (from cobalt 60) / gamma rays aimed at a tumour to destroy cancerous cells. (Use of multiple beams) May cause damage to normal cells. • radioactive inserts placed into the body to destroy cancerous cells, mainly used for prostate cancer. ignore chemotherapy, ultrasound scans, endoscopes etc.	**(6)**

Level	0	No rewardable content
1	1 - 2	• a limited description of one procedure used for either diagnosis **OR** treatment e.g. idea of tracers or an elementary notion of radiotherapy given • the answer communicates ideas using simple language and uses limited scientific terminology • spelling, punctuation and grammar are used with limited accuracy
2	3 - 4	• a simple description of one procedure used for either diagnosis **OR** treatment e.g. the tracer emits gamma rays which are detected using a gamma camera showing up area of high uptake / radioactivity; uses radioactive sources emitting beta / gamma radiation to destroy cancer cells. • the answer communicates ideas showing some evidence of clarity and organisation and uses scientific terminology appropriately • spelling, punctuation and grammar are used with some accuracy
3	5 - 6	a detailed description a procedure used for diagnosis and a procedure used for treatment. e.g. a short-lived radioactive tracer is injected into the body which then shows up areas of high activity via a gamma camera **AND** radiotherapy uses gamma to destroy cancer cells in a targeted way, with some detail. PET scanning details acceptable on the diagnosis side. (N.B. The diagnosis aspect may be covered in more detail than the treatment or vice-versa) • the answer communicates ideas clearly and coherently uses a range of scientific terminology accurately • spelling, punctuation and grammar are used with few errors

Q7.

Question Number	Answer	Acceptable answers	Mark
(i)	**C**		**(1)**

Question Number	Answer	Acceptable answers	Mark
(ii)	An explanation linking any three of the following:-		**(3)**
	positron has a positive (charge) (1)	positron has +1 / +e (charge) positron charge is +	
	electron has a {negative (charge) / opposite charge(s) } (1)	electron has -1 / -e (charge) electron charge is –	
	these charges cancel out (1)	neutralise / overall charge is zero	
	gamma rays /waves have no charge (1)	Accept for three marks: electron and positron have equal and opposite charges which cancel out.	

Question Number	Answer	Acceptable answers	Mark
(iii)	An explanation linking :		**(2)**
	positron and electron have mass(before the annihilation) (1)		
	gamma (rays produced by annihilation) have energy (1) (the equation shows)	mass (of particles) becomes energy of gamma (rays) (2)	
		all the mass before the collision becomes the energy of the gamma (rays) after the particles have been annihilated (2)	
		$E=mc^2$ reference (1) explained will get the other (1)	

Q8.

Question Number	Answer	Acceptable answers	Mark
(a)	Any one of Treatment of cancer / radiotherapy Imaging e.g.: looking at broken bones, tracers sterilizing (equipment/dressings) (1)	NOT ultrasound applications/ chemotherapy accept(to) cure/kill/detect cancer (cells) accept X-ray(s)/X-ray machine accept PET/CT scans ignore MRI scans accept (to) kill bacteria ignore medical treatment and similar vague statements	**(1)**

Question Number	Answer	Acceptable answers	Mark
(b)	12 hours = 2 half lives (1) 10 (mg) (1)	idea of halving seen e.g. 40 ÷ 2 or 20 (mg) ignore 80 (mg) and 99 ÷ 2 OR idea of 2 half lives seen or 40/4 OR (6 is 1 half-life and)12 is 2 (half-lives) OR 1/4 Give full marks for correct answer with no working.	**(2)**

Q9.

Question Number	**Answer**	**Mark**
	proton(s)	**(1)** AO 1 1

Q10.

Question Number	Answer	Acceptable answers	Mark
	An explanation linking two of the following: control rods absorb neutrons (1) control rods moved into / inserted (into reactor core) (1) to capture <u>more</u> neutrons / increase (surface) area (of control rods) (1) (and so) fewer neutrons left (to cause fission) (1)	Ignore 'to control/reduce the reaction(s)' as paraphrase of stem accept control rods take in/soak up neutrons ignore slow down neutrons accept lowered/pushed down (into reactor) NOT move(d) up AND down accept {stops / reduces number (of)} neutrons colliding (with uranium nuclei)	**(2)**